Uso dei metodi statistici per il progetto MAGIC, per la descrizione, caratterizzazione e conservazione della collezione Torraca di libri antichi, appartenenti all'Accademia pontaniana di Napoli. Use of statistical methods for the MAGIC project: application to the Accademia Pontaniana

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Launched in 2023, the Magic project stems from University of Naples "Federico II", within the field of Digital Humanities, and focuses on the establishment of a Service Center for technologies applied to cultural heritage: the first object of interest is the Francesco and Luigi Torraca collection, 186 ancient 16 th century editions housed at Accademia Pontaniana, an academy based in Naples. We present our activities focused on the digitalization, post-processing and integration into digital repositories of the digital collection we made entirely available for use on the dedicated website, www.magic.unina.it.

The process of creating the digital collection began with a preliminary analysis and inspection of the materials to be digitized. This was followed by the actual digitalization phase, the post-processing of the acquired images, and their integration into a digital object management system. The digitization was conducted using highly professional scanners able to acquire TIFF images at 600dpi, a significant value for planetary scanners. Careful page turning and attention to avoid applying any pressure, particularly on the book spines, were crucial to the successful outcome of the digitalization activities. For the long-term preservation of the obtained images, a decision was made to use the FITS format, which ensures data compatibility and durability. The digitized images were then transferred to dedicated permanent digital repositories specifically created for the collection. Informatic technology applied to philological and book sciences was used for detailed digital descriptions encompassing historical, paleographic, codicological and microbiological data, in addition to valuable physical parameters: this work covers the entire span of digitization process, metadata creation and web development of ancient books' heritage.

For the analysis of data collected for the cataloguing phase, statistics were significantly chosen to provide an overall view and synthetic vision. Starting from the catalogue of 15 th and 16 th century editions of Accademia Pontaniana edit by Silvia Sbordone, the collection of data was carried out with particular attention and meticulousness. Descriptive statistics allowed us not only to perform an exploratory analysis of the collection, but also to generate a numerical and graphical synthesis of the raw data, considering number of volumes, heterogeneity of subject, size and consistency. We are therefore prefiguring an investigation method to be applied systematically to any other collection of books. Furthermore, the results obtained from this statistical study help to better understand the characteristics: there are books of Latin literature, Roman history, Italian literature and philosophy; on the other hand, there are books of medicine, astronomy, architecture, and even esotericism. The results obtained have provided a lot of information about the collection and have also allowed us to establish rational guidelines for the protection and management of the specimens, useful for the functionality and organization of the library.

The result of the statistical study can be of support to archivists and librarians of any institution: it would be useful to characterize and preserve ancient books heritage. Based on objective data, a model comes out, functional to rationalize the digitized images in terms of calculating the average time necessary for each volume; to organize the

physical placement of the volumes themselves on the shelves by calculating for example the static load on the shelves and on the floor, or the maximum size of the books for the sizing of the shelves. It can be useful also to plan the memory space necessary on the digital repositories for the images' web diffusion.

Due to the limitation of space, the following graphs were presented in the poster, extracted from a larger number of statistic plots.

1. Frequency distribution of the year of printing of the volumes.
2. Double frequency distribution of the dedications in books by social status and gender.
3. Frequency distribution of the topics covered by the volumes.
4. Frequency histogram of the thickness of the sheets of the books.

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